

Reachability-Constrained Riemannian Planning for Cooperative Object Transport

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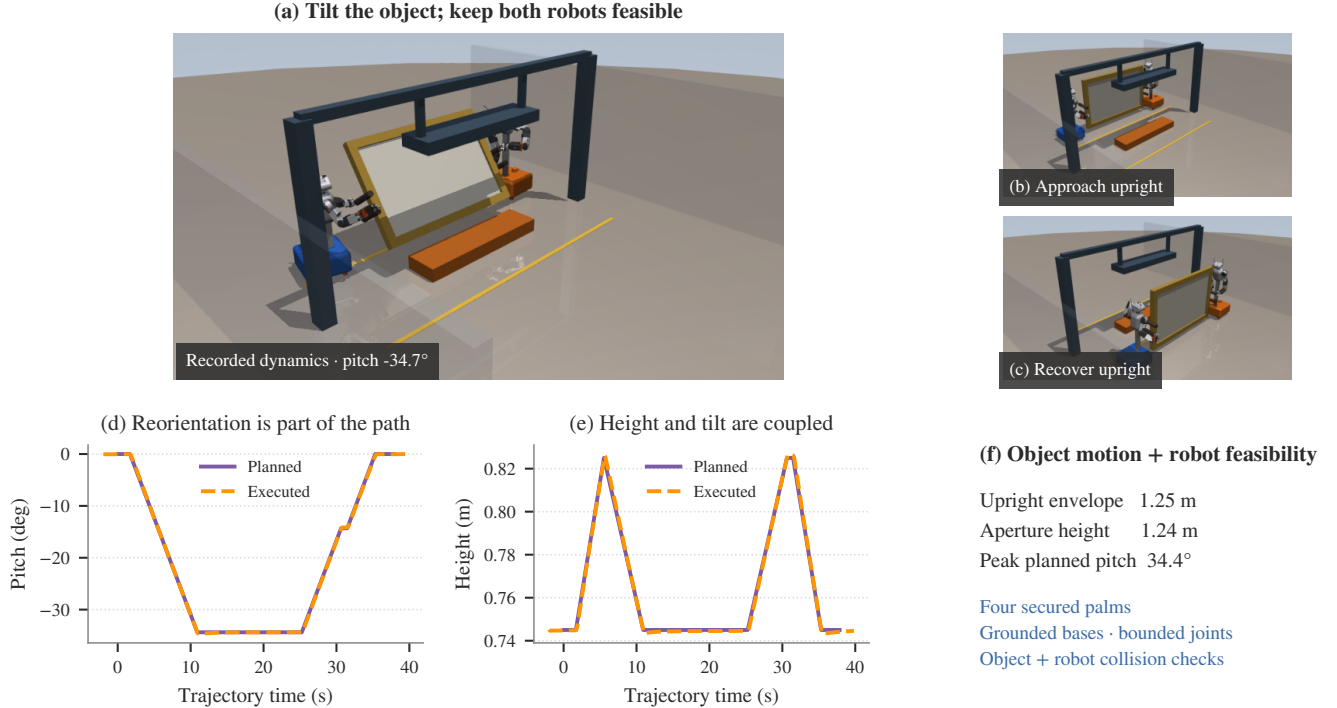


Fig. 1. Spatial cooperative transport in the museum aperture. Recorded dynamics show approach, pitch and recovery; the plots compare planned and measured object motion. The 1.25 m upright envelope exceeds the 1.24 m opening. The planner searches object pose while requiring compatible configurations of both grounded robots.

Abstract—A path for a carried object need not be executable by the robots holding it. Pitching a painting through an opening changes four palm poses, can saturate wrist or lift joints, and can bring the object into contact with its carriers. We study cooperative transport under a secure-grasp assumption. A lazy roadmap searches the object’s six-dimensional pose space using a metric induced by secured-grasp velocities. Each accepted pose retains a grounded, joint-limited configuration of both robots; connections join those same configurations and project onto the closed-chain constraints. We derive a rotation-aware product lower bound and an inertia-weighted tangent refinement, then export either four grasp-pose references or whole-body position references. The study covers nineteen existing scenes and two constructed five-room itineraries. Ten existing-scene nominal plans and both 48 m multi-room itineraries pass an independent interpolation audit; whole-body tracking reaches all ten evaluated short-scene goals. In a museum passage, a 34.4° pitch resolves a measured aperture conflict. Joint distance-time corner refinement shortens the two long routes by 6.6–9.0% and measured execution by 7.9–12.6%, while increasing actuator work. Metric ablations, 78 additional-scene trials and 138 planner benchmark trials expose sensitivity to initialization. The evidence establishes sampled feasibility in the tested models, with strong sensitivity to proposal coverage and no general optimality or energetic-superiority claim.

I. INTRODUCTION

Cooperative transport couples object motion to robot configuration. A couch may fit through a doorway after reorientation, yet a robot’s wrist may not realize the required palm attitude. Raising one end can improve obstacle clearance while exhausting the opposite robot’s lift travel. Treating the carriers as a rigid planar footprint hides these dependencies; planning only an object pose leaves them to the controller.

We consider two mobile bimanual robots carrying an already secured object (Fig. 1). Our objective is a connected sequence of object poses and compatible robot configurations, with two downstream interfaces: grasp-pose references for a task-space controller and position references for whole-body tracking. The planning problem includes object translation, roll, pitch, and yaw, while both bases remain grounded. Grasp acquisition, slip prevention, and learned coordination are outside this paper’s contribution.

Geometry-aware sampling on Riemannian manifolds supplies state-dependent metrics and local retractions [1]; matrix lower bounds can reduce the region searched after a solution is available [2]. Applying these ideas to a cooperative object

requires more than a choice of pose metric. A feasible inverse-kinematics (IK) solution at each endpoint does not establish a feasible connection, and independent IK branches can invalidate concatenated edges. We therefore retain a specific robot configuration at each roadmap vertex and validate connections between those configurations. Figure 3 shows how this object–robot coupling leads to the two execution interfaces.

The contributions are: (i) a grasp-Jacobian transport metric on $SE(3)$ coupled to bounded, grounded closed-chain IK; (ii) a lazy roadmap whose accepted connections share endpoint configurations, together with a topology-aware informed lower bound and an explicit interpolation audit; and (iii) a reproducible simulation study separating geometric planning, metric ablations, isolated pose feasibility, and downstream execution across nineteen existing scenes, two constructed five-room itineraries, and official OMPL baselines. The metric and bound are adaptations of established principles rather than a new general-purpose manifold-planning theory. Our central practical question is whether spatial object paths remain compatible with the actual carriers throughout execution.

II. RELATED WORK

Roadmaps and constrained motion. Lazy PRM postpones collision checking until candidate shortest paths require it [3]. We retain that strategy and replace an unchecked object edge with a robot-constrained local connection. Constraint-based manipulation planning, including Task Space Regions [4], and the IMACS framework [5] distinguish configuration sampling from constraint-preserving motion generation. Kingston *et al.* survey these primitives and the limitations of projection-based approaches [6]. Our method is an object-space roadmap with stored robot configurations; it does not exhaustively explore the robot configuration manifold or all grasp modes.

Riemannian planning and informed search. Kyaw and Kelly formulate sampling-based planning with state-dependent metrics and retractions [1]. Their subsequent work constructs informed regions using Loewner-order lower bounds [2]. We use a task metric induced by secured palm velocities and an isotropic product-metric lower bound on translation and rotation. Unlike a direct ellipsoid sampler in a Euclidean chart, our informed stage rejects poses using principal rotation distances. This avoids treating Euler-angle wraparound as physical distance but sacrifices direct-sampling efficiency. We do not claim geodesic optimality for our local retraction connections.

Planning and execution. The simulation uses MuJoCo’s articulated dynamics, kinematics and contact model [7]. The planner exports references independently of a particular policy. A frozen learned controller is evaluated as a downstream compatibility check, not as a learning contribution or a competing planning algorithm. All claims in this paper concern the secured transport phase; they do not extend to grasp acquisition or contact-rich pickup.

III. PROBLEM FORMULATION

Let $T = (R, p) \in SE(3)$ denote the object pose. The grounded coordinates $q = (q_1, q_2) \in \mathcal{Q}$ contain 44 variables: each robot’s base x, y, ψ , torso, arm and wrist joints, and two lift stages. Base height and inclination are fixed during planning. Fingers retain their held configuration. Joint limits define $q_{\min} \leq q \leq q_{\max}$ for the scalar articulated coordinates.

For robot $i \in \{1, 2\}$ and hand $h \in \{L, R\}$, let $A_{ih} = (R_{ih}, a_{ih})$ be the fixed object-to-palm transform measured at the held start. Secure grasp requires

$$F_{ih}(q_i) = TA_{ih}, \quad i \in \{1, 2\}, \quad h \in \{L, R\}, \quad (1)$$

where F_{ih} is forward kinematics. We use a six-dimensional residual per palm,

$$c_{ih}(q, T) = \left[\ell_R \text{Log}(R_{ih}^w(q)(RR_{ih})^\top)^\vee \right], \quad (2)$$

with $\ell_R = 0.25$ m. Two additional residuals enforce equal travel of the coupled lift stages, giving $c(q, T) \in \mathbb{R}^{26}$.

The admissible set combines closure, limits, and the configured collision model:

$$\mathcal{F} = \{(T, q) : c(q, T) = 0, \quad q_{\min} \leq q \leq q_{\max}, \quad \text{Free}(T, q)\}. \quad (3)$$

It need not be a globally smooth manifold: active joint limits, singular Jacobians, and collision boundaries create different local regimes. At a regular interior point, the implicit-function theorem applies to the equality constraints; we make no global rank assumption.

The desired output is a path $\gamma(t) = (T(t), q(t))$ from the held start to a specified object goal, with sampled evidence that it lies in $\mathcal{F}$. A grasp-reference consumer receives TA_{ih} ; a position-reference consumer receives the corresponding robot targets. Both interfaces use the same fixed grasp transforms. We deliberately distinguish this geometric problem from dynamic feasibility under actuator limits, acceleration constraints, or disturbance.

IV. REACHABILITY-CONSTRAINED RIEMANNIAN ROADMAP

A. Pose Retraction and Grasp-Induced Metric

We use world translation and angular velocity $v = (u, \omega)$ as tangent coordinates. For $T_a = (R_a, p_a)$ and $T_b = (R_b, p_b)$,

$$\Delta_{ab} = (p_b - p_a, \text{Log}(R_b R_a^\top)^\vee), \quad (4)$$

$$\text{Retr}_{(R, p)}(u, \omega) = (\text{Exp}([\omega]_\times)R, p + u). \quad (5)$$

The principal logarithm is used away from its π cut locus. This product retraction preserves valid rotations; it is not asserted to be the exponential map of the variable metric below.

The world velocity of a secured palm position is $u + \omega \times Ra_{ih}$. Consequently,

$$J_{ih}(T) = [I \quad -[Ra_{ih}]_\times], \quad (6)$$

$$G(T) = B + \sum_{i, h} J_{ih}(T)^\top W J_{ih}(T), \quad (7)$$

$$B = \text{diag}(1, 1, 1, \rho^2, \rho^2, \rho^2), \quad \rho = 0.5 \text{ m},$$

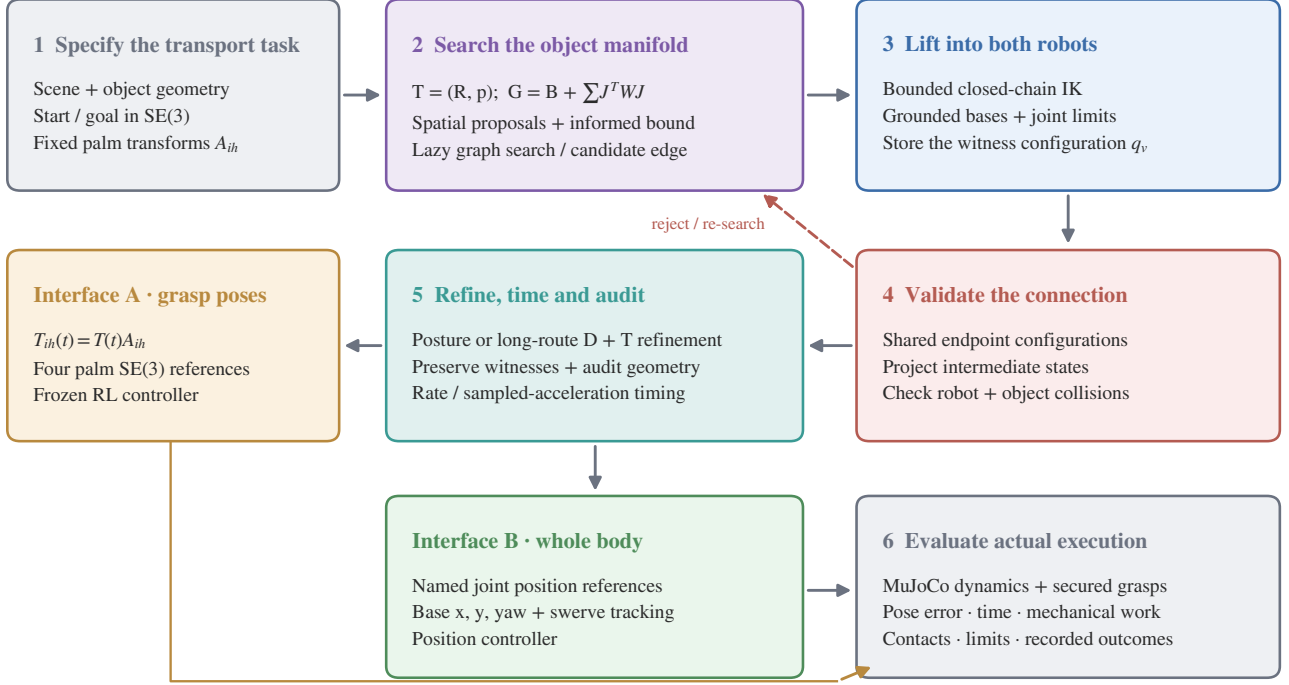


Fig. 3. System overview. A spatial object roadmap is coupled to stored robot configurations through bounded closed-chain IK. Candidate connections are checked against the full robot and object geometry; rejected connections return to graph search. The accepted configuration path supports two distinct outputs: four secured palm poses, or whole-body position references. Recorded dynamics evaluate each downstream controller separately.

with $W = \text{diag}(0.15, 0.15, 0.50)$. These weights define a transport cost and penalize vertical palm motion; they are not a calibrated energy model. The positive baseline prevents degenerate directions. For a local connection, the roadmap initially approximates length using the midpoint metric,

$$\hat{L}(T_a, T_b) = \sqrt{\Delta_{ab}^\top G(T_{1/2}) \Delta_{ab}}, \quad T_s = \text{Retr}_{T_a}(s \Delta_{ab}). \quad (8)$$

Final reported costs sum this quadrature over the densely checked path. Graph ordering remains an approximation; exact shortest paths under G are not claimed.

B. A Rotation-Aware Informed Bound

Lemma 1 (Product lower bound): Let G be given by (7), with $W \succeq 0$ and $\rho > 0$. For every piecewise smooth object path from T_a to T_b ,

$$L_G(\gamma) \geq d_B(T_a, T_b) = \sqrt{\|p_b - p_a\|^2 + \rho^2 \theta(R_b R_a^\top)^2}, \quad (9)$$

where $\theta \in [0, \pi]$ is the principal relative rotation angle.

Proof: Each $J_{ih}^\top W J_{ih}$ is positive semidefinite, so $G \succeq B$. The integrand of L_G therefore bounds the B -speed pointwise. The B -metric is the product of Euclidean translation and isotropically scaled rotation. Its minimizing rotational distance is the principal angle, and its product distance is (9). Integrating and minimizing establishes the result. ■

Thus $h(T) = d_B(T_s, T) + d_B(T, T_g)$ lower-bounds the length of any path through T . After finding a path, subsequent batches filter samples by $h(T) \leq \hat{L}_{\text{inc}}$. The inequality is exact for the continuous metric, but the implemented incumbent

uses quadrature; this filter is an informed heuristic, not an optimality certificate. No Euclidean ellipsoid is imposed on Euler angles or a global rotation-vector chart.

C. Grounded, Bounded IK and Connected Configurations

At a proposed pose, a grounded nominal formation initializes IK. Bases follow the horizontal grasp midpoints and object heading; lift stages initialize the vertical displacement. The object receives the requested full orientation, while base roll and pitch stay fixed. Each iteration solves a bounded damped least-squares subproblem,

$$\min_d \|J_c d + c\|^2 + \lambda \|d\|^2, \quad \underline{d}(q) \leq d \leq \bar{d}(q), \quad (10)$$

where $\lambda = 10^{-6}$, and the bounds combine a 0.12 per-coordinate trust box with actual joint limits. This handles active bounds inside the solve. MuJoCo's position integrator retracts the update. Nonconvergence within 70 iterations rejects the candidate; it does not prove that no other IK branch exists.

Each accepted roadmap vertex stores one configuration q_v . Candidate edges use T_s from (5), interpolate between q_u and q_v , and project each interior sample using (10). The endpoint configurations are reused exactly across incident edges. A connection is accepted only if its projected samples and inter-knot midpoint checks satisfy closure, limits, and collision predicates. This prevents an implicit jump between independently selected endpoint branches. It does not guarantee that the sampled connection is collision-free between checks.

The graph connects 18 nearest neighbors in an embedding

Algorithm 1 Lazy SE(3) roadmap with stored configurations

Require: Held (T_s, q_s) , goal T_g , bounds, fixed grasps

- 1: Initialize poses V , candidate edges E , and $C \leftarrow \infty$
 - 2: **for** each finite sample batch **do**
 - 3: Add bounded samples satisfying $h(T) \leq C$; propose neighbors
 - 4: **while** $\pi \leftarrow \text{DIJKSTRA}(V, E)$ exists **do**
 - 5: Solve bounded IK for needed vertices; store feasible q_v
 - 6: Check needed edges between their stored configurations
 - 7: **if** IK, geometry, or interpolation checks fail **then**
 - 8: Exclude the failed vertex/edge; **continue**
 - 9: **end if**
 - 10: Retain the path if its evaluated cost improves C ;
 - 11: **break**
 - 12: **end for**
 - 13: Refine robot postures and independently audit both interfaces
 - 14: **return** audited references, or a recorded failure
-

of translation and rotation matrices, supplemented by adjacent proposal layers. Dijkstra search selects a candidate route; only necessary vertices and edges are validated. Failed vertices or edges are excluded before searching again. Sampling mixes the bounded six-dimensional domain with scene-centered proposals. For five ordinary rooms, an earlier Riemannian planar roadmap supplies proposal locations; every spatial connection is still checked independently. The museum aperture uses scene-derived proposals without that planar path. This initialization is an explicit experimental prior, not evidence of unrestricted global exploration.

D. Whole-Body Refinement and Interfaces

For position tracking, we additionally refine the configuration path with a joint-centering objective f under the reduced inertia metric $H = M_q + 0.3I$. At an interior regular configuration define

$$P_H = I - H^{-1}J_c^\top(J_c H^{-1}J_c^\top)^{-1}J_c, \quad v = -P_H H^{-1}\nabla f. \quad (11)$$

Lemma 2 (Tangent descent): If $H \succ 0$ and J_c has full row rank, v in (11) minimizes $\nabla f^\top v + \frac{1}{2}v^\top H v$ subject to $J_c v = 0$. Hence $\nabla f^\top v = -v^\top H v \leq 0$.

Proof: The KKT equations are $Hv + \nabla f + J_c^\top \mu = 0$ and $J_c v = 0$. Eliminating μ yields (11). Multiplying stationarity by $v^\top$ gives the descent identity. ■

The implementation regularizes the inverse, uses short tapered steps, and reprojects each proposal. Thus the ideal lemma motivates the direction, while numerical acceptance relies on fresh constraint and collision checks. An independently audited refined path is exported; otherwise the audited roadmap configurations are retained.

For grasp tracking, interpolation occurs at the object pose before composing TA_{ih} , preserving the fixed transforms exactly. For position tracking, named joint/tendon position

targets and base x, y, ψ references are exported. A swerve tracking layer converts base references into steering positions and wheel velocities; wheel drive channels are not treated as position actuators.

E. Distance–Time Refinement for Multi-Room Routes

The roadmap cost in (8) is geometric; it does not minimize travel time. To avoid stopping at room-center turns, we replace each such turn by collision-checked quadratic curves whose formation yaw follows the tangent. Cut distances range from 1 to 3 m; the unchanged turn remains an option. Exact object poses and robot configurations are preserved at every splice. We enumerate feasible corner combinations and minimize

$$J(\gamma) = \frac{1}{2}D(\gamma)/D(\gamma_0) + \frac{1}{2}T(\gamma)/T(\gamma_0), \quad (12)$$

where D is object translation length and T is the duration under a common rate- and sampled-acceleration timing model. A forward/backward speed pass is followed by local duration increases wherever finite-difference physical accelerations exceed bounds, then a final conservative scaling check. This couples path selection to duration; it is not globally time-optimal control. Since the audited original route γ_0 remains in the finite candidate set, the selected combined objective cannot increase. Individual distance and time changes must still be measured. The room order is fixed, and no joint-centering refinement follows the timed witness export.

V. EXPERIMENTS

A. Protocol and Evidence Separation

We evaluate existing simulated environments: living room, warehouse crate, museum painting, apartment couch/doorway, workshop beam, and museum floor-clearance passage. The payload masses are 2.0, 1.6, 1.0, 2.0, 1.2, and 1.5 kg. These are simulated light-load transport tests, not evidence for carrying a full-mass household couch. Both robots begin in a secured held state. The supplied carry keyframes have approximately 0.170 m wheel clearance; we ground each base and compensate with equal lift-stage extension, preserving the initial payload and palm transforms.

For the first five scenes, the original destination x/y is retained and the goal adds 0.06 m height, 0.06 rad world roll, and 0.12 rad world pitch relative to upright. These goals exercise all spatial pose variables, but do not show that each room requires tilt to avoid an obstacle. The passage has upright start and goal, allowing geometry to determine the intermediate pitch. Five nominal runs use two batches of 180 random samples, in addition to 225 layered proposals; the passage uses one batch of 80 random samples and 510 layered proposals. Twelve additional runs use seeds 1–2 and one batch of 100 random samples. The shared planar proposal prior is preserved across comparisons.

Connection subdivisions are chosen from 0.035 m translation, 0.025 rad rotation, and 0.045 scalar-joint endpoint-difference scales. Projection and refinement can increase the actual joint increment; the largest increment in the original six-scene exports is 0.0486 in joint units. A separate audit evaluates quarter, half and three-quarter interpolated states

TABLE I

NOMINAL SE(3) PATHS AND MEASURED WHOLE-BODY EXECUTION. ALL EXPORTED PATHS PASS THE INDEPENDENT AUDIT. CLOSURE IS THE WORST INTERPOLATED PALM POSITION RESIDUAL.

Scene	Plan (s)	Closure (mm)	Goal error (mm)	WB
Living room	10.5	0.15	6.41	✓
Warehouse crate	10.7	0.12	4.23	✓
Museum painting	8.3	0.11	2.51	✓
Apartment couch	19.6	0.12	5.23	✓
Workshop beam	9.1	0.18	8.67	✓
Museum passage	13.7	0.19	1.02	✓

between all exported knots. It requires no forbidden contact deeper than 1 mm, palm position residual below 3 mm, palm angular residual below 0.02 rad, and no planning joint-limit violation. Allowed contacts are wheel–floor, secured palm–payload, and internal contacts within the same secured gripper. Other arm, robot, object and environment contacts remain disallowed. This abstraction and the compiled collision masks are part of the model.

Actual dynamics rollouts initialize state once, then advance through `mj_step`; no force is applied directly to move the payload and no state correction follows reset. Four high-impedance welds realize the assumed secure grasps. The position controller uses joint/tendon targets, gravity feedforward and swerve base feedback. Trajectory timing bounds payload translation at 0.09 m/s, rotation at 0.065 rad/s, scalar joint rates at 0.16 units/s, and base translation at 0.12 m/s. Goal success requires position error below 60 mm, angle error below 0.08 rad, and recorded forbidden penetration below 2 mm. This criterion is separate from the planning audit and does not assert exact dynamic joint-limit invariance.

B. Spatial Paths and Whole-Body Execution

All six nominal trajectories pass both interface audits. The worst inter-knot palm position residual is 0.189 mm. The additional seeded runs pass 12/12 audits. These are repeatability checks on a fixed scene set, not statistically independent samples of arbitrary homes or payloads.

Whole-body position tracking reaches 6/6 goals (Tab. I). Final position errors range from 1.02 to 8.67 mm. The greatest measured base inclination is 0.0035 rad. Soft simulated joint limits admit up to 0.0142 rad revolute excursion beyond the nominal range; no hard-limit guarantee is inferred from goal success. Recorded internal gripper overlap is excluded by the secure-grasp model and retained in the raw diagnostics.

C. Why the Museum Passage Requires Reorientation

The compiled upright payload collision envelope is 1.25 m tall, while the opening between the floor plinth and overhead edge is 1.24 m. Upright yaw and vertical translation therefore cannot fit that payload through this aperture. This local geometric statement does not exclude routes around the obstacle elsewhere in the scene.

The selected route reaches -34.38° world pitch and raises the centroid by up to 0.08 m before restoring the upright goal. Holding the object upright at the same sampled centroid

TABLE II

PAIRED METRIC ABLATION: IDENTICAL SAMPLES AND CONSTRAINTS; ALL COSTS ARE RESCORED UNDER THE PULLBACK METRIC. EACH ROW POOLS THREE SEEDS. POSITIVE REDUCTION FAVORS THE PULLBACK METRIC.

Scene	Paired success	Mean cost reduction (%)	Median time ratio
Living room	3/3	-0.002	0.98
Warehouse crate	3/3	0.242	1.01
Museum painting	3/3	0.226	0.93
Apartment couch	3/3	0.116	1.02
Workshop beam	3/3	0.592	1.01
Museum passage	3/3	0.000	0.99

positions rejects 15 configurations. The spatial route satisfies all four palm constraints and robot geometry checks. Actual whole-body execution ends with 1.02 mm position error and 0.00116 rad attitude error, with no recorded forbidden penetration. This experiment distinguishes useful intermediate reorientation from simply requesting a tilted final pose.

D. Paired Metric Ablation and Pose Feasibility

We compare the grasp pullback metric with the constant baseline B using identical pose proposals, random seeds, neighbor rules, bounded IK and collision predicates. Both variants receive one batch of 100 random samples on six scenes and three seeds (36 planned trials). Each final path is rescored under the same pullback metric; a method’s native cost is never compared directly with another metric’s units. Results in Tab. II are descriptive across 18 paired scene/seed instances. Timing includes model construction and is measured on a shared workstation; it is not an isolated throughput benchmark. The mean common-cost changes by scene range from -0.002% to 0.592% , with zero change in the passage. Thus these trials do not establish a practically large metric advantage. No population-level significance or general superiority is claimed.

We also evaluate 432 isolated poses across the six task scenes and two additional existing passages. For each scene, three height offsets, two world rotation axes and nine angles are tested. Bounded IK converges for 270 poses, but only 195 pass geometry. Thus 75 IK solutions are unsuitable under the configured collision predicate. In particular, a closed-chain solution can still place the couch through the floor or a panel against an arm. Conversely, solver nonconvergence does not establish physical unreachability.

E. Frozen Learned-Controller Interface

A frozen pre-existing SE(3) policy receives the object reference equivalent to the four grasp poses. It reaches 0/6 goals: four runs terminate under native self-penetration checks and two under base-orientation checks. These failures test interface portability, not planner feasibility or learning performance. Its self-contact interpretation differs from the secured-gripper abstraction. Checkpoint hashes, traces and videos are retained.

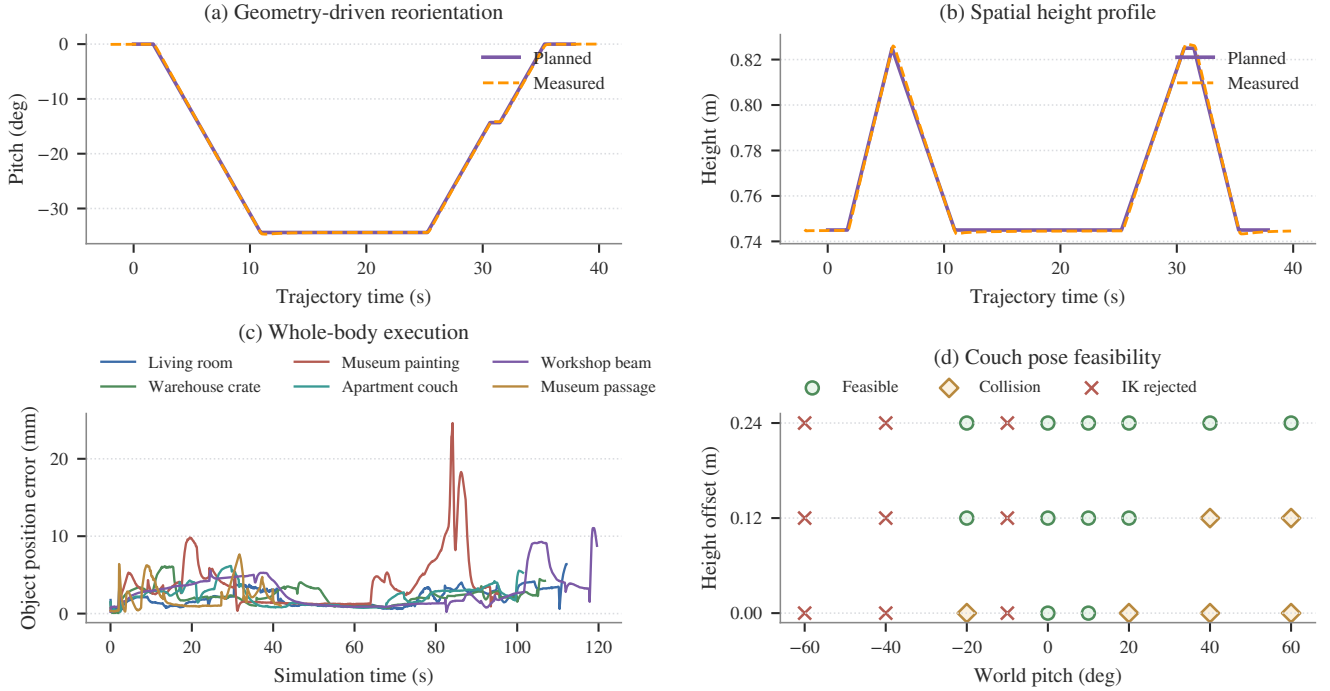


Fig. 4. Spatial planning and measured execution. (a) The museum passage’s planned and measured pitch. (b) Planned and measured object height. (c) Measured whole-body object position errors in six scenes. (d) Joint-limited, collision-checked pose probes for the living-room couch as height and world pitch vary. Pose probes are isolated configurations, not certified connections. Curves come from stored trajectories and simulator states.

F. Additional Objects and Unchanged-Scene Challenges

Thirteen additional unchanged scenes include glazing, shipping-case steps, wardrobe/soffit passages, beam tunnels, a foundry boss, tables, loading dock and painting/panel variants. Three seeds and both metrics yield 78 trials, each with 100 random samples and one batch. All failed trials are retained; finite-budget failure is not physical infeasibility. Scene-by-scene results appear in the technical notes and review site. Compiled masses define the simulated loads.

Four added nominal paths—shipping-case steps, low-beam wardrobe, foundry crate and living-room painting—pass the interpolation audit and complete whole-body tracking. The wardrobe plan reaches approximately 69° pitch. The frozen policy fails in all four added rollouts. The remaining scenes are useful negative cases: reaching a valid object pose is insufficient when the nominal roadmap cannot connect it to the start. These observations support a broader characterization of the current implementation, rather than a claim of universal object or scene generalization.

G. Five-Room Long-Horizon Planning

We construct a mirrored pair of five-room environments for a painting and a crate, with different corridor widths. Their room-center itineraries each span 48 m and traverse four corridors, with heading changes of 90° , 120° , and 90° . Repeated thresholds and overhead obstacles constrain payload height and attitude; the robot bases remain on a level floor. Corridor widths are 4.6 m and 4.8 m, respectively, accommodating the broadside payload and both carriers. These dimensions describe a cooperative formation passage,

TABLE III
FIVE-ROOM ROUTE REFINEMENT UNDER A COMMON RATE AND SAMPLED-ACCELERATION TIMING MODEL. REF. IS PREDICTED DURATION; EXEC. IS MEASURED INCLUDING PREPARATION AND SETTLING. WORK IS POSITIVE ACTUATOR MECHANICAL WORK. ORANGE MARKS THE LOWER VALUE WITHIN EACH SCENE; OURS IS THE DISTANCE–TIME REFINEMENT.

Scene	Route	D (m) ↓	Ref. (s) ↓	Exec. (s) ↓	W^+ (kJ) ↓
Museum	Room-center	48.07	631.5	635.6	2.41
Museum	Ours	44.88	581.0	585.0	8.92
Logistics	Room-center	48.41	635.4	639.5	2.47
Logistics	Ours	44.05	554.7	558.7	4.19

not a standard residential doorway.

Both complete routes pass the independent interpolation audit. Room order is supplied explicitly; each corridor and room turn uses a local spatial roadmap, reusing the previous terminal robot configuration exactly. This evaluates a prescribed itinerary, not topology discovery. The review site contains full-route maps, search traces, both exports and recordings.

We then free the three room-center corners and optimize (12). Both original and rounded routes receive the same new timing model for the comparison in Tab. III. This separates geometric savings from a change in timing assumptions. We re-execute the unchanged route with the same timing routine and controller for the paired dynamics comparison; earlier velocity-only recordings remain separate. Across 216 combinations per scene, the selected routes reduce object distance by 6.64% and 8.99%, remove in-place yaw

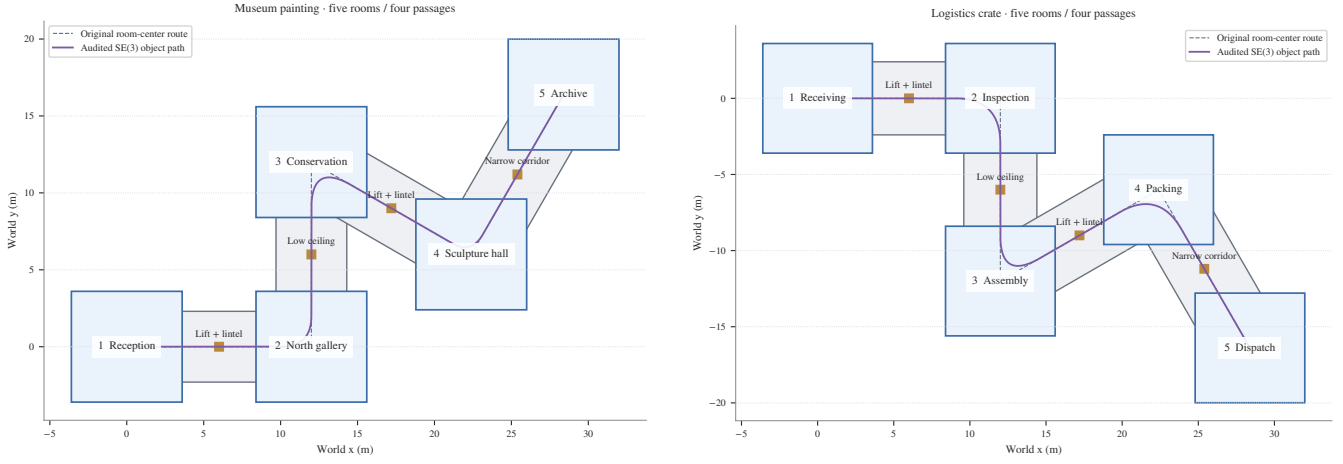


Fig. 5. Two five-room long-horizon tasks. Each prescribed itinerary traverses four passages and three large turns. Purple curves are the audited object centroid paths; the recorded 3D views additionally expose object height, attitude, both robots and full collision geometry. The layouts are generated separately from the unchanged-scene study.

turns, and complete measured tracking in 585.0 s and 558.7 s, versus 635.6 s and 639.5 s under matched baseline timing. Both reach the goal within 2.4 mm with no recorded forbidden penetration. However, positive work increases from 2.41 to 8.92 kJ and from 2.47 to 4.19 kJ. Distance–time optimization therefore does not imply energy efficiency. Candidate plots and weight sweeps are retained in the technical notes and review site.

H. External Planner Benchmark and Execution Cost

We use official OMPL 2.0.1 implementations of RRT-Connect [8], BIT* [9], AORRTC [10] and Informed RRT* [11], with the same object-pose domain, bounded closed-chain IK, robot geometry predicates and projected edge checker. Every vertex retains a specific robot witness, and every accepted solution is independently audited. Three seeds per planner and scene cover the six baseline tasks (114 trials including controls). A separate 24-trial extension tests the four OMPL planners on both five-room scenes. AORRTC returns one audited solution in 18 short-scene trials; the other external methods return none. All 24 unconditioned long-scene searches fail within the budget. These outcomes do not establish infeasibility. The long-route guided initializer additionally receives the room order and extra computation, so its success is not an isolated optimizer comparison. RRT-Connect terminates at its first exact solution; the optimizing OMPL planners use their native SE(3) path-length objectives within the solve budget. Approximate goal solutions are failures.

Scene-guided proposals are a material prior. We therefore include a uniform-sampling roadmap ablation and identify proposal-guided results separately. A deterministic prior-only control lifts the same planar itinerary with gradually changing height and attitude, validates every connection, and performs no replanning if a connection fails. It is run once per scene and tests how much of the observed success is supplied by the prior itself. All successful paths are also rescored under the common grasp-pullback metric. Feasibility and reporting units

are shared, but internal objectives and proposal distributions differ. Shared-workstation timings retain actual wall time, including native callback overruns.

For a dense path, translational travel and accumulated rotation are

$$D_p = \sum_k \|p_{k+1} - p_k\|, \quad D_R = \sum_k \|\text{Log}(R_{k+1}R_k^\top)^\vee\|. \quad (13)$$

Robot travel sums each base’s planar displacement. A common rate-limited time parameterization gives a *reference duration*; actual execution time is measured independently through the same whole-body controller on seed-0 plans, plus the successful AORRTC seed-2 case and its paired guided path. Other seeds measure planning repeatability. On the successful seed-2 museum case, AORRTC outperforms the guided path: 2.21 versus 2.31 m travel, 29.0 versus 41.9 s measured execution, and 392 versus 682 J positive work. Thus higher overall success does not imply better path quality whenever both solve the task. At each simulator step we integrate positive actuator mechanical work,

$$W^+ = \int_0^{t_f} \sum_a \max\{f_a(t)\dot{\ell}_a(t), 0\} dt, \quad (14)$$

where f_a and $\dot{\ell}_a$ are the actuator transmission force and velocity. This includes preparation and settling, excludes regeneration credit, and is not electrical energy: motor efficiency, drive losses and battery consumption are not modeled. Work from a terminated rollout is not a completed-task energy estimate. Missing rollouts receive no imputed duration or energy value. On the four baseline tasks where both the prior-only lift and guided roadmap complete seed-0 execution, the guided route reduces measured time by 0.45–4.93% and positive work by 14.21–21.49%. These are four paired controller rollouts, not a population-level energy claim. When a baseline finds no exact solution, completed-route travel and work comparisons are undefined; a higher planning success rate alone does not establish energetic superiority.

TABLE IV

PLANNER BENCHMARK WITH A 30 s SOLVE BUDGET PER TRIAL. SUCCESS INCLUDES THE INDEPENDENT INTERPOLATION AUDIT. TIME INCLUDES FAILED TRIALS; OTHER PLAN METRICS CONDITION ON SUCCESSFUL PLANS. EXECUTION AND POSITIVE ACTUATOR WORK ARE REPORTED ONLY FOR RECORDED WHOLE-BODY ROLLOUTS. THE GUIDED METHOD USES SCENE-DERIVED PROPOSALS; THE UNIFORM ABLATION REMOVES THIS PRIOR. COUNTS DISTINGUISH STOCHASTIC TRIALS AND DETERMINISTIC CONTROLS. $\uparrow/\downarrow$: HIGHER/LOWER IS BETTER. ORANGE MARKS BEST REPORTED COLUMN AGGREGATES (TIES INCLUDED); SUCCESSFUL-SCENE SUBSETS DIFFER. OURS DENOTES THE SCENE-GUIDED ROADMAP.

Planner	Success $\uparrow$	Time (s) $\downarrow$	Travel (m) $\downarrow$	Rotation (rad) $\downarrow$	Reference (s) $\downarrow$	Execution (s) $\downarrow$	Work (J) $\downarrow$
OMPL RRT-Connect	0/18	30.37	—	—	—	—	—
OMPL BIT*	0/18	30.40	—	—	—	—	—
OMPL AORRTC (2025)	1/18	30.32	2.21	1.07	24.95	28.98	391.97
OMPL Informed RRT*	0/18	30.33	—	—	—	—	—
Uniform roadmap	0/18	9.73	—	—	—	—	—
Prior-only lift	4/6	2.94	7.11	3.18	108.64	112.67	907.42
Ours	18/18	5.32	6.74	3.17	101.05	103.42	681.63

VI. DISCUSSION AND CONCLUSION

The experiments support a restricted conclusion: object-space SE(3) planning can produce connected, joint-limited cooperative configurations and executable position references in the tested secured-grasp simulations. The museum experiment shows a case where intermediate pitch resolves a measured vertical aperture conflict while preserving robot compatibility.

Several limitations bound that conclusion. The graph stores one IK branch per pose and uses proposal priors; alternative branches and homotopy classes can be missed. Collision checks are discrete and tolerate submillimetre penetration rather than enforcing a positive clearance margin. The metric is a designed grasp-motion cost, not measured energy, and approximate edge lengths do not provide an optimality guarantee. Grasp acquisition, frictional retention, actuator-limited dynamic planning and full-mass payloads are not evaluated. The fixed scene collection, constructed room itineraries and three planning seeds do not establish broad generalization or hardware readiness. The nine unsuccessful added scenes are evidence that roadmap coverage and the chosen IK branch remain substantial limitations.

The resulting planner remains separate from learned-controller design. Its two interfaces expose the same secured-object geometry at different levels of detail. The OMPL benchmark separates native search behavior from the strong scene-proposal prior. Longer compute budgets, shared adaptive proposals, diverse grasp configurations and independent constrained-state-space implementations remain necessary before broad planner rankings are justified. Continuous collision validation and physical transport require separate evidence.

The long-route distance–time refinement reduces both travel and measured duration on the two constructed layouts, but

increases positive actuator work. A future objective can include work or torque estimates explicitly; the present results do not support an energy-minimization claim.

REFERENCES

- [1] P. T. Kyaw and J. Kelly, “Geometry-aware sampling-based motion planning on Riemannian manifolds,” in *Workshop on the Algorithmic Foundations of Robotics*, 2026, arXiv:2602.00992, version 3. [Online]. Available: <https://arxiv.org/abs/2602.00992v3>
- [2] —, “Direct informed sampling on Riemannian manifolds via Loewner order lower bounds,” *arXiv preprint arXiv:2606.02879*, 2026. [Online]. Available: <https://arxiv.org/abs/2606.02879>
- [3] R. Bohlin and L. E. Kavraki, “Path planning using lazy PRM,” in *IEEE International Conference on Robotics and Automation*, vol. 1, 2000, pp. 521–528.
- [4] D. Berenson, S. Srinivasa, and J. Kuffner, “Task space regions: A framework for pose-constrained manipulation planning,” *The International Journal of Robotics Research*, vol. 30, no. 12, pp. 1435–1460, 2011.
- [5] Z. Kingston, M. Moll, and L. E. Kavraki, “Exploring implicit spaces for constrained sampling-based planning,” *The International Journal of Robotics Research*, vol. 38, no. 10–11, pp. 1151–1178, 2019.
- [6] —, “Sampling-based methods for motion planning with constraints,” *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 1, pp. 159–185, 2018.
- [7] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A physics engine for model-based control,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033.
- [8] J. J. Kuffner and S. M. LaValle, “RRT-Connect: An efficient approach to single-query path planning,” in *IEEE International Conference on Robotics and Automation*, 2000, pp. 995–1001.
- [9] J. D. Gammell, S. S. Srinivasa, and T. D. Barfoot, “Batch informed trees (BIT*): Sampling-based optimal planning via the heuristically guided search of implicit random geometric graphs,” in *IEEE International Conference on Robotics and Automation*, 2015, pp. 3067–3074.
- [10] T. Wilson, W. Thomason, Z. Kingston, and J. D. Gammell, “AORRTC: Almost-surely asymptotically optimal planning with RRT-Connect,” *IEEE Robotics and Automation Letters*, vol. 10, no. 12, pp. 13 375–13 382, 2025.
- [11] J. D. Gammell, S. S. Srinivasa, and T. D. Barfoot, “Informed RRT*: Optimal sampling-based path planning focused via direct sampling of an admissible ellipsoidal heuristic,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2014, pp. 2997–3004.